

Exercise 3

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Problem 1a.

Problem 1b.

Proof. $\forall a, d, b, c \in \mathbb{Z} \ni ad - bc = 1$ and $\forall z \in \mathcal{H}$.

Note the following, $\forall (m, n) \in \mathbb{Z} \setminus \{(0, 0)\}$

$$\begin{aligned} \left(\frac{1}{m \left(\frac{az+b}{cz+d} \right) + n} \right)^2 &= \left(\frac{1}{\frac{m(az+b)+n(cz+d)}{cz+d}} \right)^2 \\ &= \left(\frac{cz+d}{z(ma+nc) + (mb+nd)} \right)^2 \end{aligned}$$

Analogously

$$\frac{1}{\left| m \left(\frac{az+b}{cz+d} \right) + n \right|^{2\epsilon}} = \frac{|cz+d|^{2\epsilon}}{|z(ma+nc) + (mb+nd)|^{2\epsilon}}$$

Thus,

$$\begin{aligned} G_{2,\epsilon} \left(\frac{az+b}{cz+d} \right) &= \frac{1}{2} \sum_{(m,n) \neq (0,0)} \left(\frac{1}{m \left(\frac{az+b}{cz+d} \right) + n} \right)^2 \frac{1}{\left| m \left(\frac{az+b}{cz+d} \right) + n \right|^{2\epsilon}} \\ &= \frac{1}{2} (cz+d)^2 |cz+d|^{2\epsilon} \sum_{(m,n) \neq (0,0)} \frac{1}{|z(ma+nc) + (mb+nd)|^{2\epsilon}} \frac{1}{(z(ma+nc) + (mb+nd))^2} \\ &= \frac{1}{2} (cz+d)^2 |cz+d|^{2\epsilon} \sum_{(u,v) \neq (0,0)} \frac{1}{|uz+v|^{2\epsilon}} \frac{1}{(uz+v)^2} \\ &= (cz+d)^2 |cz+d|^{2\epsilon} G_{2,\epsilon}(z) \end{aligned}$$

Where the 2nd last equality comes from the fact that $\begin{bmatrix} a & c \\ b & d \end{bmatrix} \in SL_2(\mathbb{Z})$ is an invertible linear transformation which takes every nonzero lattice element of $\mathbb{Z}^2$ to a unique nonzero lattice element of $\mathbb{Z}^2$. $\square$

Problem 2a.

Proof. 1. $Ix = x$:

$$Ix = \frac{1x}{1} = x$$

2. $\forall a_i, b_i, c_i, d_i \in \mathbb{Z}$ for $i \in [1, 2]$ where $a_id_i - b_ic_i = 0$. $\forall x \in P^1(\mathbb{Q})$

$$\begin{aligned} \left(\begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \right) (x) &= \begin{bmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{bmatrix} (x) \\ &= \frac{x(a_1a_2 + b_1c_2) + a_1b_2 + b_1d_2}{x(c_1a_2 + d_1c_2) + c_1b_2 + d_1d_2} \\ &= \frac{a_1(a_2x + b_2) + b_1(c_2x + d_2)}{c_1(a_2x + b_2) + d_1(c_2x + d_2)} \\ &= \frac{a_1 \frac{a_2x+b_2}{c_2x+d_2} + b_1}{c_1 \frac{a_2x+b_2}{c_2x+d_2} + d_1} \\ &= \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \left(\frac{a_2x + b_2}{c_2x + d_2} \right) \\ &= \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix} \left(\begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} x \right) \end{aligned}$$

Thus the mobius transform is a valid action on $P^1(\mathbb{Q})$ □

Problem 2b.

Proof. It suffices to show that $\forall x \in P^1(\mathbb{Q})$, $\exists g \in SL_2(\mathbb{Z}) \ni g \cdot 0 = x$.

1. $0 \rightarrow \infty$: Let $g = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \implies g \cdot 0 = \frac{-1}{1 \cdot 0} = \infty$

2. $0 \rightarrow x$ for $x \in \mathbb{Q} \setminus \{0\}$: $x = \frac{p}{q}$ for $p, q \in \mathbb{Z} \setminus \{0\} \ni (p, q) = 1$. Thus $\exists m, n \in \mathbb{Z} \ni mp - nq = 1$. Let

$$g = \begin{bmatrix} -n & p \\ -m & q \end{bmatrix} \in SL_2(\mathbb{Z}) \implies g \cdot 0 = \frac{-n(0)+p}{-m(0)+q} = \frac{p}{q}$$

3. Note $0 \rightarrow 0$ is trivial since the mobius transform is an action.

Thus the action of $SL_2(\mathbb{Z})$ on $P^1(\mathbb{Q})$ is transitive since everything in $P^1(\mathbb{Q})$ is in the orbit of 0.

TODO: Proof for finite orbits is trivial use finite index of Γ . □